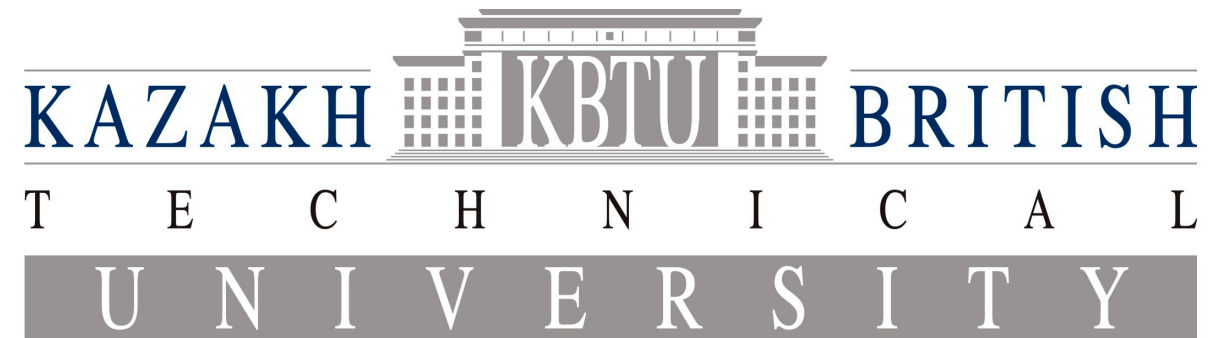




WEB DEVELOPMENT

Lesson 11

Django Rest Framework Simple JWT

<https://django-rest-framework-simplejwt.readthedocs.io/en/latest/>

What is JWT?

What is a JSON Web Token?

A JSON web token, or JWT (“jot”) for short, is a standardized, optionally validated and/or encrypted container format that is used to securely transfer information between two parties.

What is a JSON Web Token?

JWTs consist of three parts separated by dots (.), which are:

- header
- payload
- signature

JWT typically looks like the following:

```
xxxxx.yyyyy.zzzzz
```

Header

The header typically consists of two parts: the type of the token, which is JWT, and the hashing algorithm such as HMAC SHA256 or RSA.

```
{  
  "alg": "HS256",  
  "typ": "JWT"  
}
```

this JSON is Base64Url encoded to form the first part of the JWT.

Payload

The second part of the token is the payload, which contains the claims. Claims are statements about an entity (typically, the user) and additional metadata.

```
{  
  "sub": "1234567890",  
  "name": "KBTU SITE",  
  "admin": true  
}
```

The payload is then Base64Url encoded to form the second part of the JWT.

Signature

To create the signature part you have to take the encoded header, the encoded payload, a secret, the algorithm specified in the header, and sign that.

```
HMACSHA256(  
  base64UrlEncode(header) + "." +  
  base64UrlEncode(payload),  
  secret)
```

The signature is used to verify that the sender of the JWT is who it says it is and to ensure that the message wasn't changed in the way.

Putting all together

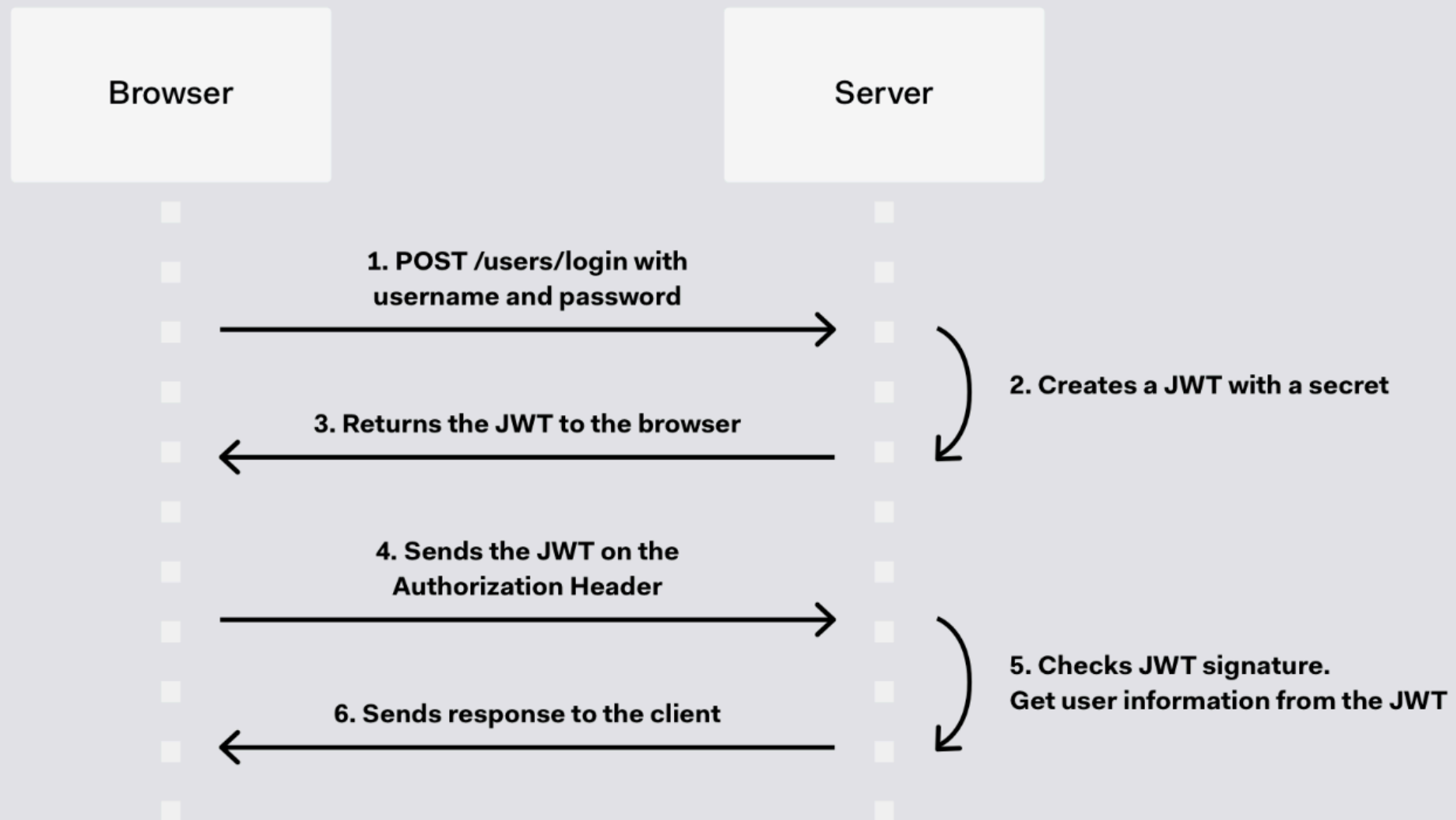
The following shows a JWT that has the previous header and payload encoded and it is signed with a secret.

```
eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.  
eyJzdWIiOiIxMjM0NTY3ODkwIiwibmFtZSI6IkpvaG4  
gRG9lIiwiaXNTb2NpYWwiOi0nRydWV9.  
4pcPyMD09o1PSyXnrXCjTwXyr4Bsezdi1AVTmud2fU4
```

Access to resource

header of each http request must contain

Authorization: JWT/Bearer <token>



Pagination

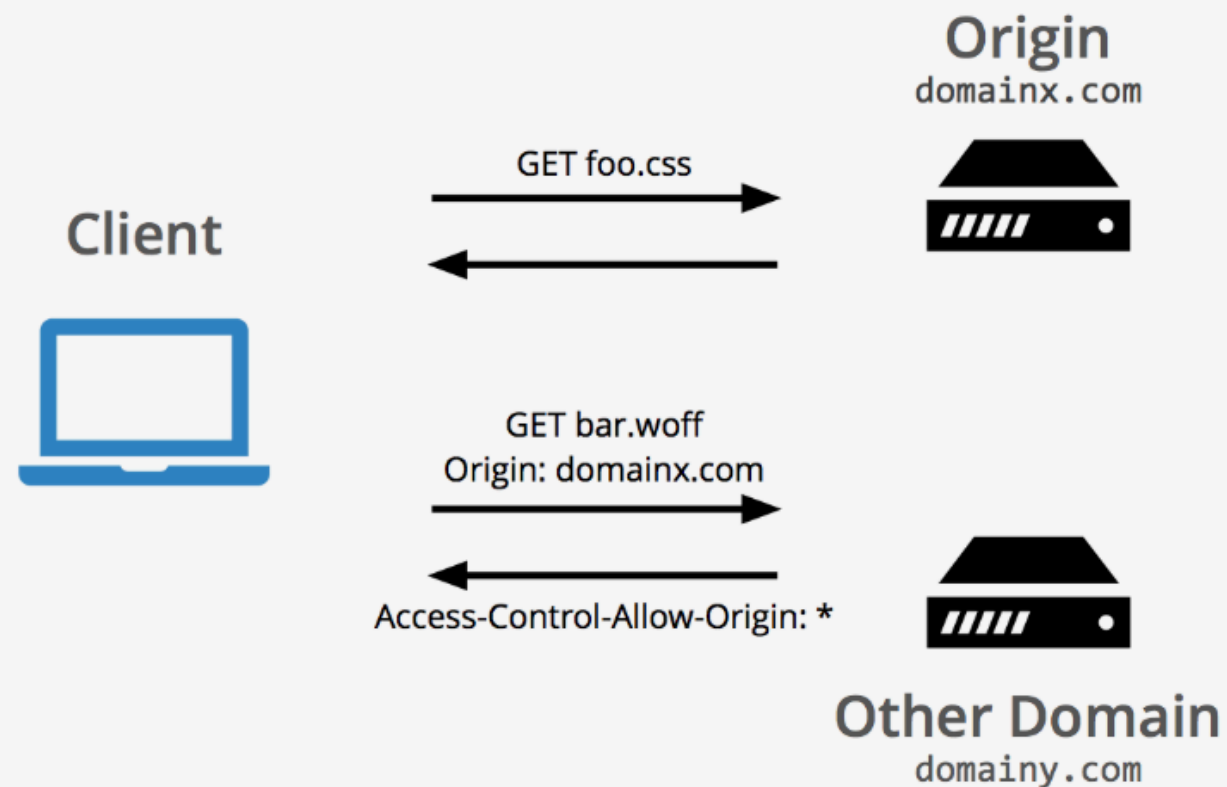
Pagination

```
REST_FRAMEWORK = {  
    'DEFAULT_PAGINATION_CLASS': 'rest_framework.pagination.PageNumberPagination',  
    'PAGE_SIZE': 10  
}
```

Cross-Origin Resource Sharing (CORS)

<https://github.com/ottoyiu/django-cors-headers>

Cross-Origin Resource Sharing (CORS)



CORS

Cross-Origin Resource Sharing (CORS)

✖ Access to Font at '...' from (index):1
origin '...' has been blocked by CORS policy: No 'Access-Control-Allow-Origin' header is
present on the requested resource. Origin '...' is therefore not allowed access. The response
had HTTP status code 404.

Questions?